Radner Equilibrium: Definition and Equivalence with Arrow-Debreu Equilibrium, Asset Markets and No Arbitrage

Econ 3030 Fall 2025

Lecture 26

Outline

1. Sequential Trade and Arrow Securities
2. Radner Equilibrium
3. Equivalence between Arrow-Debreu and Radner Equilibria
4. Asset Markets and No Arbitrage

Arrow-Debreu Equilibrium in the One Good Timeless Exchange Economy

- One commodity, no firms, one future period, and no consumption today;
  - dated events are $(1, \{s\})$, so we label everything by the realized state $s \in S$.
- A consumption bundle is an S-dimensional vector $\mathbf{x}_i \in X_i \subset \mathbb{R}_+^S$ and

$$\mathbf{x}_i \succsim \mathbf{y}_i \quad \text{if and only if} \quad \sum_{s=1}^{S} \pi_{si} u_i(x_{si}) \geq \sum_{s=1}^{S} \pi_{si} u_i(y_{si})$$

Definition

$\mathbf{x}^* \in \mathbb{R}_+^{S \times I}$ and $\mathbf{p}^* \in \mathbb{R}_+^S$ are an Arrow-Debreu equilibrium if

1. For every i, $\mathbf{x}_i^* \in \mathbb{R}_+^S$ solves: $\max_{\mathbf{x}_i \in \mathbb{R}_+^S} \sum_{s=1}^{S} \pi_{si} u_i(x_{si})$ subject to $\sum_{s=1}^{S} p_s^* x_{si} \leq \sum_{i=1}^{I} p_s^* \omega_{si}$
2. Markets clear: $\sum_{i=1}^{I} x_{si}^* \leq \sum_{i=1}^{I} \omega_{si}$ for $s = 1, \ldots, S$

Timing of Trades in Arrow-Debreu

Remark

In an Arrow-Debreu economy, all decisions are made at date 0.

- Individuals exchange promises to deliver and receive quantities of the goods according to the realized dated-event.
  - They trade now for all future deliveries.
- As time and uncertainty unfold, these promises are carried through exactly as planned, without changes.
- What if tomorrow there were also markets for the L goods?
- Is there an incentive to trade in these spot markets?

Spot Markets Do Not Matter in Arrow-Debreu

Fact

Given an Arrow-Debreu equilibrium, there are no incentives to trade in spot markets.

Proof.

Suppose not: consumers can find a mutually beneficial trade in some dated-event.

- Thus, in that dated-event there exist a feasible allocation that consumers prefer to the equilibrium bundle, with at least one consumer preference being strict.
  - consumers trade only if they can do better.
- This new allocation is feasible, and Pareto dominates the Arrow-Debreu equilibrium.
- But this is impossible, because the First Welfare Theorem holds. □

Spot and Forward Markets

Remark

Most real world markets are spot, not forward.

Objective

Reconcile this observation with a general equilibrium model with uncertainty similar to Arrow-Debreu.

Idea

- The main role of state contingent commodities is to allow welfare transfers across states...
- If one can have achieve that same objective with just a few dated-event contingent commodities,
- then only some forward markets are needed, and most trade takes place in spot markets.

Radner Equilibrium

- Individuals make decisions now and in the future
  - only one commodity is traded now and in the future
    - this commodity can be used to transfer wealth across states,
  - all other commodities are traded only in markets that open after uncertainty is resolved.
- Today's decisions depend on forecasts about the future: trade is sequential, and expectations are crucial.
- This equilibrium concept is called Radner equilibrium.

Main Idea

- Replace forward markets with expectations about future spot markets.
- Given these, consumers decide about welfare transfers across states at date 0.
- Afterwards, consumers trade in markets for physical goods.

Main Result: Radner and Arrow-Debreu Are Equivalent

If expectations are correct, and individuals have some effective way to transfer wealth across states, a Radner equilibrium is equivalent to an Arrow-Debreu equilibrium: having fewer markets does not matter.

Sequential Trading and Arrow Securities

Date 0

Only one physical good is used as a state-contingent commodity: money.
Amounts of money are traded for delivery if and only if a particular state occurs.

- These are called Arrow securities.

Date 1

Some state occurs, and markets open where commodities trade at current prices.

- Individuals trade decisions depend on (i) current prices, and (ii) their income;
  - income depends on their trades in the state-contingent commodity that corresponds to the realized state.

Expectations Are Crucial

Date 0 trades reflect what consumers think will happen in the future markets.

- To decide how much to trade of the state-contingent commodity individuals make consumption plans for each possible state; these plans depend on their forecasts of the future spot prices.

Sequential Trading and Arrow Securities in an Exchange Economy

Notation

- $\mathbf{z}_i = (z_{1i}, .., z_{Si}) \in \mathbb{R}^S$ denotes i's trades in the state-contingent commodity;
  - this *Arrow security* specifies amounts to be delivered, or received, of commodity 1 in each state.
    - **note**: these trades can be negative or positive.
- $\mathbf{q} = (q_1, .., q_S) \in \mathbb{R}^S_+$ denotes the prices of the Arrow security.
- $\mathbf{x}_i = (\mathbf{x}_{1i}, .., \mathbf{x}_{Si}) \in \mathbb{R}^{L\times S}_+$ denotes i's consumption plans vector;
  - $\mathbf{x}_{si} = (x_{1si}, .., x_{Lsi}) \in \mathbb{R}^L_+$ denotes i's expected consumption in state s;
    - **note**: $\mathbf{x}_{1i} = (x_{11i}, .., x_{1Si}) \in \mathbb{R}^S_+$ represents expected trade in commodity 1, the commodity for which there are also date 0 markets.
- $\mathbf{p} = (\mathbf{p}_1, .., \mathbf{p}_S) \in \mathbb{R}^{L\times S}_+$ is the vector of expected prices;
  - $\mathbf{p}_s \in \mathbb{R}^L_+$ is the expected price vector for the L goods in state s.

- **Remark**: $\mathbf{x}_i$ and $\mathbf{p}$ are expected: they represent planned choices and prices.
- Consumers make choices and plans given current and expected prices.
  - There is no date 0 consumption (for simplicity).

Optimization

Given current prices ($\mathbf{q}$) and expected prices ($\mathbf{p}$), each individual maximizes utility.

- A consumption plan includes current trades in the state-contingent commodity ($\mathbf{z}_i$) and expected purchases in each possible state ($\mathbf{x}_i$).

Consumers' Choices at time 0

At date 0, consumer i solves the following maximization problem

$$\max_{\mathbf{z}_i \in \mathbb{R}^S, \mathbf{x} \in \mathbb{R}_+^{L \times S}} U_i(\mathbf{x})$$

subject to $\underbrace{\mathbf{q} \cdot \mathbf{z}_i \leq 0}_{\text{budget constraint at time 0}}$ and $\underbrace{\mathbf{p}_s \cdot \mathbf{x}_{si} \leq \mathbf{p}_s \cdot \boldsymbol{\omega}_{si} + p_{1s} z_{si} \quad \text{for each } s}_{\text{expected budget constraints at time 1}}$

- There will be S budget constraints at date 1: from date 0 perspective, these constraints are "in expectation": $\mathbf{x}_{si}$ is expected consumption; $\mathbf{p}_s$ are expected prices.
- **Remark**: at date 1 good 1 enters in three parts:
  - buy the desired amount for immediate consumption (x_{1si}),
  - sell the endowment (ω_{1si}), and
  - sell the realized state-contingent "outcome" of date 0 trades (z_{si}).

Time 0 Budget Constraint

Date 0 budget constraint

$$\mathbf{q} \cdot \mathbf{z}_i = \sum_{s=1}^{S} q_s z_{si} \leq 0$$

- The individual has no income at time 0, hence her trades cannot have positive cost.
  - Since the price vector is non-negative, if she promises to buy good 1 in state s ($z_{si} > 0$), she must also promise to sell it in some other state t ($z_{ti} < 0$).
  - She cannot promise to spend more than she makes
    - one could add a positive date 0 endowment (and date zero consumption), without affecting this logic.
- The time 0 budget constraint is homogeneous of degree zero in prices.
  - Therefore, we can normalize time 0 prices.

Time 1 Expected Budget Constraints

Date 1 expected budget constraints

$$\sum_{l=1}^{L} p_{ls} x_{lsi} \leq \sum_{l=1}^{L} p_{ls} \omega_{lsi} + p_{1s} z_{si} \qquad \text{for each } s = 1, ..., S$$

- In each state, the individual expects to trade in the markets that correspond to that state; her expected wealth reflects the outcome of time 0 trades.
- We can normalize by assuming that $p_{s1} = 1$ for each s.
  - Thus, Arrow security promises to receive/deliver units of money (good 1).
- Since there are no "sign" restrictions on $\mathbf{z}_i$, we may have $z_{si} < -\omega_{1si}$ in some state s:
  - i promises to deliver more than she will have, and her income must compensate.
    - the ability to do this is limited because consumption cannot be negative.

- Arrow securities allow wealth transfers across states: at date 0, a consumer can buy a state s dollar and pay for it with a state t dollar.
  - If state s occurs, she uses the extra dollar to buy commodities;
  - if state t occurs, she has one less dollar to buy commodities.

Radner Equilibrium

In equilibrium, everything happens exactly as planned.

Definition

A Radner equilibrium is composed by state-contingent commodities prices $\mathbf{q}^* \in \mathbb{R}^S$, trades $\mathbf{z}_i^* \in \mathbb{R}^S$, spot prices $\mathbf{p}_s^* \in \mathbb{R}^L$ for each s, and consumption $\mathbf{x}_i^* \in \mathbb{R}^{L \times S}$ for each i such that:

1. for each individual, $\mathbf{z}_i^*$ and $\mathbf{x}_i^*$ solve

$$\max_{\mathbf{z}_i \in \mathbb{R}^S, \mathbf{x}_{si} \in \mathbb{R}^L_+} \sum_{s=1}^S \pi_{si} u_{si}(\mathbf{x}_{si}) \quad \text{subject to}$$

$$(i) : \sum_{s=1}^S q_s^* z_{si} \leq 0$$

$$(ii) : \mathbf{p}_s^* \cdot \mathbf{x}_{si} \leq \mathbf{p}_s^* \cdot \boldsymbol{\omega}_{si} + p_{1s}^* z_{si} \quad \text{for all } s$$

2. all markets clear:

$$\sum_{i=1}^I \mathbf{z}_{si}^* \leq 0 \qquad \text{and} \qquad \sum_{i=1}^I \mathbf{x}_{si}^* \leq \sum_{i=1}^I \boldsymbol{\omega}_{si} \qquad \text{for all } s$$

- Markets clear at consumption plans that maximize individuals' utility, given current and expected prices.

Radner and Arrow-Debreu Are Equivalent

- The timing of decisions is unimportant.

Proposition (Equivalence of Arrow-Debreu and Radner equilibria)

1. *Let the allocation* $\mathbf{x}^* \in \mathbb{R}^{L \times S \times I}$ *and the prices* $\mathbf{p}^* \in \mathbb{R}^{L \times S}_{++}$ *constitute an Arrow-Debreu equilibrium. Then, there are prices* $\mathbf{q}^* \in \mathbb{R}^{S}_{++}$ *and state-contingent commodity trades* $\mathbf{z}^* = (\mathbf{z}_1^*, .., \mathbf{z}_I^*) \in \mathbb{R}^{S \times I}$ *such that:* $\mathbf{z}^*$, $\mathbf{q}^*$, $\mathbf{x}^*$, *and* $\mathbf{p}_s^*$ *for each* s *form a Radner equilibrium.*
2. *Let* $\mathbf{x}^* \in \mathbb{R}^{L \times S \times I}$, $\mathbf{z}^* \in \mathbb{R}^{S \times I}$, $\mathbf{q}^* \in \mathbb{R}^{S}_{++}$ *and* $\mathbf{p}_s^* \in \mathbb{R}^{L}_{++}$ *for all* s *form a Radner equilibrium. Then, there are* S *strictly positive numbers* $\mu_1, .., \mu_S$ *such that the allocation* $\mathbf{x}^*$ *and the state-contingent commodities price vector* $(\mu_1 \mathbf{p}_1^*, .., \mu_S \mathbf{p}_S^*) \in \mathbb{R}^{L \times S}_{++}$ *form an Arrow-Debreu equilibrium.*

1. An Arrow-Debreu equilibrium can be made into a Radner equilibrium by appropriately choosing the state contingent commodity trades and prices.
2. A Radner equilibrium can be made into an Arrow Debreu equilibrium by appropriately modifies spot prices into state-contingent prices.

- The proof only needs to show that the two budget sets are the same.

The Importance of Arrow Securities

The propositon says that Arrow-Debreu equilibrium and Radner equilibrium are equivalent, provided there are enough Arrow securities.

Arrow-Debreu and Radner are equivalent

- An Arrow-Debreu equilibrium can be made into a Radner equilbrium.
- A Radner equilibrium can be made into an Arrow-Debreu equilibrium.

- Then the timing of trades is not that important.
  - Equilibrium outcomes are the same if we imagine all trade taking place at time 0, or if we imagine that Arrow securities are traded at that time and the other trades occur in spot markets.
- Having all the state-contingent commodities trades is not that important.
  - Equilibrium outcomes are the same if we imagine a full set of state-contingent commodities, or if we only imagine one Arrow security per state.